RESEARCH POSTER / AI-ASSISTED SECURITY OPERATIONS

Wazuh Security Response Copilot

SERC: retrieval-augmented guidance for incident response

Ismail et al. | Sensors 2025, 25, 870 | Published 31 January 2025

WHAT EXACTLY IS THIS WORK ABOUT?

The **Security Event Response Copilot (SERC)** adds an AI assistant to Wazuh. It extracts the meaning of security alerts, retrieves relevant incident-response and cybersecurity guidance, and asks an LLM to produce tailored mitigation instructions that analysts can review and discuss in the dashboard.

01 INPUTS

WAZUH SECURITY EVENT

A JSON alert collected from monitored endpoints.

Useful fields

- Alert description
- Severity level
- Event / log details
- Compliance tags
- MITRE ID, when present

THREE KNOWLEDGE SETS

1. Incident-response guides and playbooks
2. NIST CSF 2.0
3. MITRE ATT&CK tactics, techniques and mitigations

PREPARE THE CORPUS

Split long guides into coherent text chunks. Embed and store content in three Qdrant collections; preserve complete use-case documents where appropriate.

02 METHODOLOGY

1 REFINE THE ALERT

Parse the Wazuh JSON log. A Pixtral-based extractor uses a one-shot prompt to summarize the issue and key metadata.

Output: a structured, readable event summary

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2 ENCODE FOR RETRIEVAL

Use BAAI/bge-large-en-v1.5 to represent text as vectors. Search Qdrant for semantically similar guidance.

Similarity measure: cosine similarity

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3 ADD SECURITY CONTEXT

Retrieve incident-response, NIST and MITRE context. When a MITRE ID is available, use it for a direct MITRE lookup.

Combine retrieved guidance with event details

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4 GENERATE RESPONSE GUIDANCE

A prompt builder combines the alert and retrieved context. The LLM produces tailored instructions. **Compared models: GPT-4o and Pixtral 12B.**

03 OUTPUTS

INCIDENT-RESPONSE INSTRUCTION GUIDE

Context-specific actions to investigate the event and mitigate the identified risk, presented in Markdown.

FRAMEWORK CONTEXT

Relevant MITRE tactics and techniques, with guidance informed by incident-response references and NIST CSF 2.0.

WAZUH COPILOT TAB

Recommendations appear inside the dashboard. Follow-up chat supports clarifications about the incident.

PRACTICAL PURPOSE

Help SOC analysts interpret alerts and decide how to respond.

04 EVIDENCE FROM THE PAPER

12 simulated attack scenarios | Recommendations compared with SOCFortress reference playbooks

Model	BERTScore F1 range	Peak cosine similarity
Pixtral 12B	85.4-87.2%	71.69%
GPT-4o (OpenAI)	84.9-86.5%	75.81%

Atomic Red Team generated endpoint activity; Wazuh collected the resulting alerts.

These are response-text similarity scores, not attack-detection accuracy or verified mitigation success.

05 APPLICATION AREAS

SOC incident handling: contextual guidance for alerts.
Analyst assistance: explanations and follow-up chat.
Security exercises: review responses to account compromise, malware, phishing and ransomware scenarios.

RESEARCH INTERPRETATION

The contribution connects Wazuh alerts to retrieved response knowledge. Text similarity does not verify that advice is correct or safe to execute. The study does not establish response-time savings or field effectiveness; SOAR automation and fine-tuning are future work.

Source: Ismail et al., “Enhancing Security Operations Center: Wazuh Security Event Response with Retrieval-Augmented-Generation-Driven Copilot,” Sensors 25 (2025), 870.

DOI: 10.3390/s25030870 | Sections 3-5; Figures 9-12. Model names and numerical ranges follow the paper. Results are author-reported, not independently reproduced.